HISTORICAL RECREATION DOCUMENTATION

Push Engraving on Copper Plate:
The Conversion on the Way to Damascus

Period-Accurate Materials, Tools, Techniques, Process, and Sources
For Presentation at Historical Reenactment Events

Prepared April 2026

1. Object Scope and Historicity

1.1 Description of the Object

This project is a hand-pushed burin engraving on a hammered copper plate depicting the biblical scene of Saul's conversion on the road to Damascus (Acts 9:1–9). The engraving is executed using period-accurate tools, materials, and techniques consistent with the practice of European goldsmith-engravers of the 15th–16th centuries. The finished plate serves as both a decorative engraved object and a functional intaglio printing plate from which paper impressions can be pulled.

1.2 Skills Required

This project requires the integration of multiple distinct skills: metallurgy and tool-making (forging, hardening, and tempering steel gravers from high-carbon tool steel); woodturning (turning graver handles from dense hardwood on a lathe); metalwork (hammering and polishing the copper plate); draftsmanship (converting a painted source image into a reversed line drawing suitable for engraving); design transfer (applying the reversed drawing to the polished plate using period transfer methods); push engraving (cutting lines, hatching, cross-hatching, and stippling with a burin using hand pressure only); sharpening (maintaining precise cutting geometry on natural stones); and optionally, intaglio printing (inking, wiping, and pressing the plate to produce paper impressions).

1.3 The Biblical Subject

The Conversion of Saul on the Road to Damascus is recorded in Acts 9:1–9, with parallel accounts in Acts 22:6–11 and Acts 26:12–18. Saul of Tarsus, a persecutor of the early Christian community, is traveling to Damascus with letters authorizing the arrest of Christians. Near the city, a brilliant light from heaven strikes him to the ground and he hears a voice: "Saul, Saul, why do you persecute me?" Blinded, Saul is led into Damascus, where after three days his sight is restored and he is baptized, becoming Paul the Apostle.

This scene was a popular subject for 16th-century artists. The National Gallery of Art notes that "from a religious perspective, the image of Christ winning over Saul represented divine power and specifically the victory of the Church over its enemies" and that "for artists, the story provided an opportunity to show off their skills in what often came to be treated as a battle scene."

> **Primary Source:** *Acts 9:1–9, Acts 22:6–11, Acts 26:12–18 (New Testament)*
>
> **Secondary Source:** *Italian Paintings of the Sixteenth Century: The Conversion of Saint Paul, c. 1544, National Gallery of Art, Washington*

1.4 Historical Exemplars

The following are documented, extant historical exemplars demonstrating that push engraving of biblical scenes on copper was a well-established practice throughout the 15th and 16th

centuries. These exemplars span multiple geographic regions, artistic traditions, and decades, providing compelling evidence that this project replicates an authentic period craft.

Exemplar 1: Martin Schongauer, Passion Series (c. 1475–1480)

A series of twelve burin engravings on copper depicting scenes from the Passion of Christ. Schongauer (c. 1450–1491) was the son of a Colmar goldsmith and the most important printmaker north of the Alps before Dürer. His 116 surviving engravings were distributed across Europe. The Passion series exemplifies the practice of engraving biblical narrative scenes for devotional use.

> **Location:** *British Museum, London; Victoria and Albert Museum, London; Metropolitan Museum of Art, New York; and other major print collections worldwide*
>
> **Source:** *V&A Museum Blog: "The Resurrection comes from a series of twelve engravings of The Passion of Christ, which could perhaps have been purchased as single sheets, or as the whole series, depending on one's desires or one's budget."*

Exemplar 2: Albrecht Dürer, The Three Master Engravings (1513–1514)

Three burin engravings on copper: Knight, Death and the Devil (1513), St. Jerome in His Study (1514), and Melencolia I (1514). These represent the highest technical achievement of pre-1600 push engraving. Dürer (1471–1528) learned burin technique from his goldsmith father and perfected it over decades. The engravings demonstrate the full expressive range of the burin — line, hatching, cross-hatching, and stippling — applied to complex figurative and symbolic compositions.

> **Location:** *Held in numerous major print collections worldwide, including the Metropolitan Museum of Art, British Museum, and Rijksmuseum*
>
> **Source:** *Wikipedia, "Albrecht Dürer": "During the same period Dürer perfected the difficult art of using the burin to make engravings. Most likely he had learned this skill during his early training with his father, as it was also an essential skill of the goldsmith."*

Exemplar 3: Maso Finiguerra, Niello Paxes (c. 1452–1464)

Engraved silver paxes with niello fill, produced for the Florence Baptistery. Finiguerra (1426–1464) was a Florentine goldsmith celebrated by Benvenuto Cellini as "the finest master of his day in niello engraving." His work demonstrates that engraving of religious scenes on metal (for liturgical rather than printing purposes) was standard goldsmith's practice in 15th-century Italy. Sulfur casts and paper impressions taken from his engraved plates survive, documenting the practice of recording engraved designs.

> **Location:** *Niello paxes in the Museo Nazionale del Bargello, Florence; sulfur casts in the British Museum; niello plaques at the Metropolitan Museum of Art*
>
> **Source:** *Britannica, "Maso Finiguerra": "He also made niello prints, which are impressions from engraved silver on paper. Examples of his sulfur casts and niello prints are still in existence."*

Exemplar 4: Hans Baldung Grien, The Conversion of St. Paul, woodcut (c. 1515–17)

A single-sheet woodcut depicting the same biblical subject as this project. The composition shows Paul and his horse fallen, two mounted companions, and Christ with a tau cross in the

clouds. While a woodcut rather than an engraving, this demonstrates that the Conversion of Paul was an established subject for printmakers in the early 16th century. The bold composition and dramatic treatment of the scene provide a period model for the subject.

> **Location:** *British Museum, London (Inv. 1895,0122.303)*
>
> **Source:** *British Museum collection record: "The Conversion of St Paul; St Paul and his horse on the ground, struck and blinded by the bright light which is set against a dark sky and sent by Christ."*

Exemplar 5: Lucas van Leyden, The Conversion of St. Paul, engraving

A burin engraving on copper depicting the same subject. Lucas van Leyden (1494–1533) was one of the leading engravers of the Low Countries. This exemplar is directly relevant because it is both an intaglio burin engraving and depicts the identical biblical scene, demonstrating that the Conversion of Paul was an established subject for copper-plate engravers.

> **Location:** *Metropolitan Museum of Art, New York (Acc. 41.1.28)*

Additional Evidence of Use and Purpose

Engraved devotional prints served documented purposes in the 15th–16th centuries: as affordable alternatives to painted devotional images for middle-class buyers; as personal devotional aids that could be inserted into prayer books; as commercial products sold individually or in series; and as pattern sources for other artists working in painting, sculpture, and other media. The V&A Museum notes that "printed images on paper were to some extent inexpensive alternatives to paintings" and that Schongauer's prints "could perhaps have been purchased as single sheets, or as the whole series, depending on one's desires or one's budget." Goldsmiths also engraved directly onto liturgical objects (chalices, paxes, reliquaries) for ecclesiastical use, and onto secular objects (armor, jewelry, belt fittings) for decorative purposes.

> **Source:** *V&A Museum Blog, "Twenty Objects for Twenty Years: Martin Schongauer"*
>
> **Source:** *Wikipedia, "Old Master Print": "Serving as a pattern for artists may have been a primary purpose for the creation of many prints."*

2. Materials

Every material used in this project is discussed below with its historical basis, specific period evidence, and — where modern substitutions are made — a comparative analysis explaining the deviation and its justification.

2.1 The Plate: Copper

Historical Material

Copper was the universal plate material for intaglio engraving from the 1430s through the end of the period and beyond. It was chosen because it is soft enough to cut cleanly with hand tools, hard enough to hold fine detail, does not rust, and can produce many impressions before wearing. Period copper plate was produced by hammering ingots or sheet to the desired thickness and then polishing to a mirror finish.

> **Primary Source:** *Copper Development Association, "This History of Intaglio": "Copper was the preferred metal because it was strong yet easily worked and did not rust."*

Period vs. Modern Copper

Period copper plate was hammered (not rolled). Modern commercially available copper sheet is cold-rolled, producing a different grain structure. Andrew Rafferty, a modern engraver who attempted to replicate Dürer's technique, found that "modern copper plate is rolled and has a grain, unlike the copperplate of the late 15th century which was hammered" and that "the curved cross-hatching that gives Dürer's work its unique solidity is less difficult to achieve on hammered plate."

> **Source:** *TheHistoryofArt.org, "Albrecht Dürer Engravings"*

Decision

For this project, the copper plate is hand-hammered from copper sheet and polished to a mirror finish, replicating the period preparation method. If modern cold-rolled copper sheet is used as a starting point, this substitution is justified by availability and cost — the functional difference is in grain structure only, and the rolled sheet can be further worked with a planishing hammer to approximate hammered plate. The material itself (copper) is identical to the period material.

2.2 Tool Steel: Gravers

Historical Material

Period gravers were made from high-carbon steel, hardened and tempered by the engraver or a toolsmith. Theophilus Presbyter's De Diversis Artibus (c. 1100–1120), the oldest surviving complete treatise on metalworking crafts, describes the making and heat treatment of steel tools for engraving and related work in Book III. Commercially available gravers in the period would have been forged from bloomery or crucible steel with a carbon content sufficient to harden by quenching — roughly equivalent to modern W-1 (water hardening) or O-1 (oil hardening) tool steel, which are plain high-carbon steels.

> **Primary Source:** *Theophilus Presbyter, De Diversis Artibus (c. 1100–1120), Book III. English translation: Hawthorne and Smith, On Divers Arts (Dover, 1979)*

Comparative Analysis: A1 Tool Steel vs. Period Steel

This project uses A1 tool steel, an air-hardening tool steel. A1 differs from period carbon steel in that it contains chromium and molybdenum additions that allow it to harden in air rather than requiring a liquid quench. The functional result — a hardened steel cutting tool at approximately Rockwell C 58–62 — is identical. The advantages of A1 for this application are: reduced risk of quench cracking on the thin cross-sections of gravers; minimal distortion during heat treatment; and the engraver's existing familiarity with the material from prior toolmaking experience (chasing chisels and repoussé punches).

The substitution is justified on practical grounds: the cutting performance, hardness, and edge-holding characteristics are functionally equivalent to period carbon steel. The tool's behavior under the burin is indistinguishable from a period graver. For maximum period authenticity, O-1 or W-1 tool steel could be used instead, as these are plain carbon steels closer in composition to period bloomery steel, requiring oil or water quench respectively. Old file steel is another documented traditional source material that closely approximates period carbon steel.

> **Source:** *Engraver's Cafe forum, "Making Homemade Engravers": Experienced engravers report making functional gravers from O-1, W-1, drill rod, and old file steel, all of which are plain high-carbon steels.*
>
> **Source:** *Engraver's Cafe forum, "Making My Own Gravers": "I've made gravers out of O-1 which is fine for soft metal engraving."*

2.3 Graver Handles: Dense Hardwood

Historical Material

Period graver handles were turned from dense hardwood, most commonly boxwood (Buxus sempervirens). The handle is mushroom-shaped with one side cut flat, approximately 1.5–2 inches (4–5 cm) in length, providing an ergonomic palm grip that positions the shaft low to the work surface.

> **Source:** *Grokipedia, "Burin (engraving)": "The handle is typically mushroom-shaped and constructed from wood (such as boxwood), measuring approximately 1.5–2 inches (4–5 cm) in length."*
>
> **Source:** *1911 Encyclopaedia Britannica, "Line-Engraving": "A handle rather like a mushroom with one side cut away."*

Decision

Handles for this project are turned on a lathe from boxwood or equivalent dense hardwood (hornbeam, dogwood, or similar). This is identical to the period material. Each graver has its own dedicated handle with the tang friction-fitted into a drilled hole — no ferrule, no set screw. This replicates the documented period practice in which engravers maintained a set of individually handled tools.

2.4 Sharpening Stones

Historical Material

Period sharpening was done exclusively on natural stones. Arkansas stones (novaculite) and other fine-grained natural oilstones were the standard. A leather strop with polishing compound was used for final finishing.

> **Source:** *Ganoksin, "Metal Engraving Tools": "Grinding is usually done on an oilstone, working in a circular stroke, first on a coarse stone and then on a finer one. Follow this with a few strokes on a hard leather with a polishing powder to create a mirror bright finish."*

Decision

Natural Arkansas stones and a leather strop with polishing compound are used for this project. No diamond plates, ceramic stones, or powered sharpening systems are used. This is identical to period practice.

2.5 Design Transfer Medium

Historical Method

Period engravers transferred designs by tracing from a paper drawing using a stylus. The transfer medium was soot or lamp black applied to the back of the drawing, which left faint guide lines on the polished plate when traced over with a pointed tool.

> **Source:** *Grokipedia, "Burin (engraving)": "The design is transferred to the plate by lightly marking outlines with a drypoint needle or tracing from a paper drawing using carbon paper or a stylus."*
>
> **Source:** *Curtidos Cabezas: "Transfer: print your initials and transfer with carbon paper or tracing."*

Decision

Lamp black (soot from a candle or oil lamp) rubbed onto the back of the reversed drawing, then traced with a steel stylus. Modern carbon paper is a functional equivalent but not strictly period; lamp black transfer is the documented period method and is used for this project.

2.6 Ink and Paper (If Printing)

If paper impressions are pulled from the finished plate: period printing ink was oil-based, made from lampblack ground in linseed oil. Period paper was handmade laid paper (showing chain lines and laid lines from the papermaking mould), dampened before printing. Wove paper was not invented until the 18th century and would be anachronistic. Modern handmade laid paper is available from specialty suppliers and is functionally identical to period paper.

3. Methods and Tools

Every method and tool used in this project is described below with its historical basis. Where any deviation from strictly period practice occurs, it is identified, described, justified, and compared to the period method.

3.1 Tool Making

Forging the Gravers

The graver blanks are forged from A1 tool steel bar stock to a square or lozenge cross-section, approximately 3–4mm across and 100–115mm (4–4.5 inches) long. The forging is done at a coal or gas forge, drawing out the bar to the desired profile and tapering the tang end to fit the handle. This replicates the period practice of goldsmiths making their own engraving tools, as documented by Theophilus and described by later sources.

> **Primary Source:** *Theophilus, De Diversis Artibus, Book III: describes making and heat-treating steel tools for metalwork*
>
> **Source:** *Engraver's Cafe, "Making Homemade Engravers": French engraver describes forging gravers from old file steel, heat treating at 850°C, quenching in water, and fitting boxwood handles turned on a lathe.*

Heat Treatment

The forged blanks are hardened by heating to non-magnetic temperature (approximately 1475°F / 800°C for carbon steels; higher for A1 due to its alloy content) and air-cooling (A1) or quenching in oil or water (O-1 / W-1). They are then tempered to a straw color to achieve approximately Rockwell C 58–62. This hardness range allows the tool to hold a sharp edge while cutting soft metals (copper, brass, silver) without being so brittle that the tip chips readily. Period heat treatment used the same principles — heating to critical temperature and quenching — with color judgment as the primary means of gauging temperature, exactly as practiced here.

> **Source:** *Ganoksin: "Gravers are sold in their hardened state and any grinding should be done in such a way that this temper is preserved. Touch the tool lightly to the grinding wheel, dipping it frequently into water to prevent overheating. If the steel starts to show color the temper has been compromised."*

Sharpening Geometry

The cutting face is ground at 45° on the diagonal, with 15° heels (relief bevels on the underside). For hand pushing specifically, a lower face angle of 35–40° with a longer heel is preferred, as this geometry requires less force and provides more control than the steeper angles used with modern pneumatic tools. Sharpening is done on natural Arkansas stones with oil, followed by a leather strop. The face and heel surfaces must be flat and polished to a mirror finish.

> **Source:** *Antique Firearms Restoration Blog, "Graver Sharpening": "The main face is cut at 45 degrees on the diagonal and the heels are cut at 15 degrees. These angles can be varied according to personal preferences, but 45/15 are a good starting point."*
>
> **Source:** *Engraver's Cafe, "Graver Geometry for Hand Pushing": experienced hand-push engravers recommend lower face angles of 35–40° and longer heels for hand work.*

Handle Turning

Handles are turned on a lathe from boxwood or equivalent dense hardwood to the documented period mushroom-with-flat-bottom shape. A hole is drilled at a slight angle from the flat face to receive the tang. The tang is friction-fitted by tapping the handle onto the tang held in a vise. Period engravers used this identical method.

3.2 Plate Preparation

The copper sheet is cut to size, hammered to working thickness if needed (period plates were hammered, not rolled), and polished to a scratch-free mirror finish. Period polishing was done progressively with finer abrasives and burnishing. For this project, the plate is polished using progressively finer grades of abrasive, finishing with a polishing compound — functionally identical to period practice, though modern abrasive papers may be used in place of period abrasive stones as a convenience substitution with no functional difference.

3.3 Design Preparation and Transfer

The source image (a pre-1600 painting of the Conversion of Paul) is studied and interpreted as a line drawing. This is a creative act of translation — the engraver must decide how to represent painted tonal values as line work (hatching, cross-hatching, stippling). This process of converting paintings into engraved line compositions was standard period practice; Marcantonio Raimondi (c. 1480–1534) built his career on engraving reproductions of Raphael's paintings, and Britannica notes that "by the 16th century, the reproductive role of engraving had become so firmly established that Italy's greatest master of engraving technique, Marcantonio Raimondi, is mainly known for his copies of Raphael's paintings."

> **Source:** *Britannica, "Engraving"*

The line drawing is reversed (since the print will be a mirror image of the plate) and transferred to the polished plate using lamp black on the back of the drawing, traced with a steel stylus. Dürer is documented as having made preparatory drawings in reverse to plan his engravings.

> **Source:** *TheHistoryofArt.org: "Dürer made preparatory drawings, in reverse, to plan out the fine lines of the cross hatching, as a preparatory drawing still exists in Vienna."*

3.4 Engraving Technique

Grip and Cutting

The burin is held at approximately 30° to the plate surface. The mushroom handle is cradled in the palm and the index and middle fingers guide the shaft. Pressure is applied both downward and forward. The tool cuts into the metal, forming a chip and pushing a thin curl of metal ahead of it. This is push engraving — no hammer, no pneumatic assist, hand pressure only.

> **Source:** *Wikipedia, "Burin (engraving)": "The burin is held at approximately 30° to the surface. The index and middle fingers guide the shaft, while the handle is cradled in the palm."*

Plate Rotation

The plate rests on a leather sandbag cushion and is rotated with the non-cutting hand to produce curves. Schongauer "was the first engraver to curve parallel lines, probably by rotating the plate against a steady burin." The Diderot Encyclopédie plates clearly illustrate the plate resting on a small turntable cushion.

> **Source:** *Wikipedia, "Martin Schongauer"*

Line Vocabulary

All tonal variation is achieved through line character and density. The line vocabulary consists of: pure line (continuous clean strokes for contours); hatching (parallel lines of varying spacing and depth for shading); cross-hatching (crossed sets of parallel lines for deeper shadows, pioneered by Schongauer and perfected by Dürer); and stippling (dots and short flicks for texture, documented from the late 15th century). The Metropolitan Museum describes Schongauer's technique: he "used a great variety of strokes — from the long, sinuous lines that create the beard of the saint and the curling fur of one of the demons, to the short flecks of the saint's coarsely woven robe; he also made use of crosshatching in the deepest shadows to model the forms."

> **Source:** *Wendy Thompson, "The Printed Image in the West: Engraving," Met Museum*

Workflow Sequence

The period engraving workflow, documented across multiple sources: (1) design preparation; (2) plate preparation and polishing; (3) design transfer in reverse; (4) contour engraving (main outlines first); (5) shading (hatching, cross-hatching); (6) textures and details (stippling, fine work); (7) cleaning and finishing (removing burrs and metal curls); (8) printing if desired (inking, wiping, pressing).

> **Source:** *Curtidos Cabezas, "Engravings with Burin": "Burin engraving process: design → transfer → securing → main cuts → shading → cleaning and finishing → protection."*

3.5 Deviations from Period Methods

The following deviations from strictly period practice are made in this project, with justification:

A1 tool steel instead of period carbon steel: Justified by the engraver's existing familiarity with this material from prior toolmaking. The functional result (hardened steel graver at Rc 58–62) is identical. See Section 2.2 for full comparative analysis.

Modern abrasive papers for plate polishing: Used as a convenience alongside traditional polishing methods. The functional result (mirror-polished copper surface) is identical to period preparation. Period engravers used progressively finer natural abrasive stones and burnishing to achieve the same result.

Electric lighting: Period engravers worked by natural light or candlelight. Modern lighting is used for practical reasons at indoor events. This has no effect on the object itself.

No pneumatic or power-assisted tools are used. No diamond or ceramic sharpening equipment is used. No modern quick-change handle systems are used. The engraving technique itself — push engraving with a hand-held burin — is identical to period practice in every respect.

4. Research and Supporting Evidence

4.1 Types of Evidence

This project's documentation draws on multiple types of evidence:

Primary written sources: Period treatises by practicing craftsmen describing tools, materials, and techniques from firsthand experience. These include Theophilus Presbyter's De Diversis Artibus (c. 1100–1120), Benvenuto Cellini's Trattati dell'oreficeria (1560s), and Abraham Bosse's Traité des manières de graver en taille douce (1645, documenting practices unchanged from the 16th century).

Extant objects: Surviving engraved copper plates, prints pulled from those plates, engraved niello silverwork, and engraved decorative metalwork held in major museum collections (Metropolitan Museum of Art, British Museum, Rijksmuseum, Victoria and Albert Museum, Bargello). These provide direct physical evidence of period materials, techniques, and subjects.

Visual documentation: Abraham Bosse's engraved illustrations of the engraver's workshop (1643–45), the Diderot Encyclopédie plates on "Gravure en taille-douce" (1751–80, documenting practices unchanged from the 16th century), and period depictions of goldsmith workshops.

Modern scholarly analysis: Museum publications, art historical scholarship, and technical analyses of period engravings. Key sources include the Metropolitan Museum's Heilbrunn Timeline of Art History essays on engraving, the National Gallery of Art's exhibition catalogs, and Britannica's scholarly articles.

Practitioner knowledge: Modern hand engravers who practice period techniques, documented in specialized forums (Engraver's Cafe, Engraving Forum) and instructional materials (Sam Alfano, GRS courses). While modern, these practitioners preserve and transmit technical knowledge about hand-tool geometry, sharpening, and cutting technique that is directly continuous with period practice.

4.2 Discussion of Sources and Historic Context

The practice of push engraving on copper for the production of devotional prints emerged from the European goldsmithing trade in the 1430s, as documented by Britannica: "Engraving originated independently in the Rhine valley in Germany and in northern Italy about the middle of the 15th century. It seems to have been first developed by German goldsmiths." The Wikipedia article on Goldsmiths confirms: "The printmaking technique of engraving developed among goldsmiths in Germany around 1430, who had long used the technique on their metal pieces. The notable engravers of the fifteenth century were either goldsmiths, such as Master E.S., or the sons of goldsmiths, such as Martin Schongauer and Albrecht Dürer."

The Italian tradition developed simultaneously from niello work. The Wikipedia article on Niello records that "at least in Italy, some of the very earliest engraved prints were in fact made by treating a silver object intended for niello as a printing plate with ink, before the niello was added." Britannica notes that Maso Finiguerra (1426–1464) "also made niello prints, which are

impressions from engraved silver on paper. Examples of his sulfur casts and niello prints are still in existence."

The earliest and most important primary source on medieval metalworking tools and techniques is Theophilus Presbyter's De Diversis Artibus. The Britannica entry on Theophilus confirms it is "an exhaustive account of the techniques of almost all the known crafts of the first half of the 12th century" written by a Benedictine monk who was also a practicing metalworker, believed to be Roger of Helmarshausen, whose surviving metalwork demonstrates the techniques described in the text.

Abraham Bosse's 1645 Traité des manières de graver en taille douce is the first published treatise specifically on engraving. While published just after the period boundary, it documents practices and tools that were functionally unchanged from the previous century, as the burin, the sandbag cushion, the sharpening methods, and the printing process it illustrates are all attested in earlier sources and extant objects. The treatise was widely translated and remained the standard reference for over a century.

5. Execution, Complexity, and Creativity

5.1 Project Goals

The following goals were established for this project:

1. Forge and heat-treat a set of push engraving gravers from tool steel, replicating the period practice of engravers making their own tools.

2. Turn period-accurate mushroom handles from dense hardwood.

3. Prepare a copper plate using period-appropriate methods.

4. Convert a pre-1600 painting of the Conversion of Paul into a reversed line drawing suitable for engraving, exercising personal artistic interpretation in the translation from painted tonal values to line work.

5. Transfer the design and engrave the plate using push engraving only, employing the full period line vocabulary (line, hatching, cross-hatching, stippling).

6. Produce a finished engraved plate that functions both as a decorative object and as a printing plate capable of producing paper impressions.

7. Document every material, tool, and technique decision with reference to period sources.

5.2 Personal Interpretation

This project is not a direct copy of an existing engraving. The source material is a painting, and the conversion of a painted image into engraved line work requires substantial personal artistic interpretation. The engraver must decide: which painted values become lines and which become negative space; how to represent color and tonal gradation using only black-and-white line vocabulary; how to simplify or adapt the composition for the constraints of the burin; and how to manage the reversal of the image. This process of converting paintings to engravings was itself a recognized and respected skill in the period — Marcantonio Raimondi was celebrated specifically for his ability to translate Raphael's paintings into copper plate engravings.

5.3 Challenges and Solutions

Challenge 1: Learning Push Engraving as a New Skill

Push engraving requires a fundamentally different hand motion from the engraver's prior experience with chasing and repoussé (hammer-driven) and woodblock carving (knife-driven). The burin is pushed with palm pressure, requiring development of new muscle memory for control of line depth, width, and curvature. Solution: extensive practice on scrap copper with straight lines, curves, and basic scroll elements before beginning the finished plate, following the progression recommended by experienced engravers.

Challenge 2: Converting Painted Tonal Values to Line Work

A painting has continuous tonal gradation; an engraving has only lines. Every area of the composition must be interpreted as some combination of line density, and there is no way to correct major compositional decisions once cutting has begun. Solution: extensive preparatory drawing in pencil, working through the tonal interpretation on paper before committing to the plate. Multiple studies at different simplification levels to determine the appropriate level of detail for the engraver's current skill.

Challenge 3: Image Reversal

The engraved image must be cut in reverse so that it prints correctly. This requires the engraver to work with a mirrored version of the design, which can be disorienting. Solution: prepare the reversed drawing carefully before transfer, and work from the transferred lines on the plate rather than from memory or a non-reversed reference.

Challenge 4: Sharpening to Precise Geometry

A burin that is not perfectly sharpened with correct face and heel geometry will dig in, skip, or produce ragged lines. Maintaining consistent geometry during resharpening (which is required frequently) is critical. Solution: develop a consistent sharpening routine using the natural stone and strop, with the thumbnail test to verify sharpness before each session. Practice resharpening until it becomes reliable and fast.

Challenge 5: Tool Making with Precision Geometry

While the engraver has prior experience making chasing tools, gravers require more precise geometry at the cutting face. The face angle, heel angle, and point geometry must all be correct for the tool to function properly. Solution: forge multiple blanks and practice grinding geometry on natural stones before committing to the final tool set.

5.4 Future Avenues of Exploration

This project opens several avenues for further development:

Niello fill: Adding niello (a sulfur-silver-copper-lead compound) to the engraved lines to create a high-contrast decorative object in the Florentine goldsmith tradition. This would add a new skill (niello preparation and application) and connect the project to the Italian engraving tradition documented by Finiguerra, Cellini, and Theophilus.

Intaglio printing: Building or obtaining a simple screw press and pulling editions of paper prints from the plate. Period presses were simple wooden constructions, and a period-accurate press is itself a viable reenactment project.

Engraving directly on liturgical or decorative objects: Moving from flat plates to three-dimensional objects such as chalice bands, paxes, or belt fittings, replicating the goldsmith's decorative practice that predated printmaking.

Sulfur casting: Making sulfur casts from the engraved plate to record the design, replicating the documented practice of Finiguerra and other Renaissance goldsmiths.

More complex compositions: Progressing to multi-figure scenes with full cross-hatched shading in the manner of Dürer's master engravings.

Period pigment preparation: Making printing ink from lampblack ground in linseed oil, and preparing laid paper, to produce fully period-accurate prints.

6. Bibliography

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